

NAE12

November 20, 2022

```
[ ]: import numpy as np #
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
```

```
[ ]:
```

```
x_1=[4.0825,4.0725,3.9318,3.7050,3.5050,3.4619,3.6568,4.0700,4.5712,4.9602,3.  
↳5825,3.5725,3.4318,3.2050,3.0050,2.9619,3.1568,3.5700,4.0712,4.4602]  
y_1=[4.0565,4.1864,4.2922,4.2702,4.0706,3.7345,3.3899,3.2031,3.3045,3.7206,3.  
↳5565,3.6864,3.7922,3.7702,3.5706,3.2345,2.8899,2.7031,2.8045,3.2206]  
x_2=[2.0825,2.0725,1.9318,1.7050,1.5050,1.4619,1.6568,2.0700,2.5712,2.9602,2.  
↳5825,2.5725,2.4318,2.2050,2.0050,1.9619,2.1568,2.5700,3.0712,3.4602]  
y_2=[2.0565,2.1864,2.2922,2.2702,2.0706,1.7345,1.3899,1.2031,1.3045,1.7206,2.  
↳5565,2.6864,2.7922,2.7702,2.5706,2.2345,1.8899,1.7031,1.8045,2.2206]
```

```
X = np.zeros((2*len(x_1),2))  
Y =_  
↳[1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,  
for i in range(len(x_1)):  
    X[i,0] = x_1[i]  
    X[i,1] = y_1[i]  
    X[len(x_1)+i,0] = x_2[i]  
    X[len(x_1)+i,1] = y_2[i]
```

```
[ ]: import matplotlib.pyplot as plt      # 8  
plt.figure()                             #  
X1 = np.array(x_1)                       # X  
Y1 = np.array(y_1)                       # Y  
plt.scatter(X1,Y1,s=25,color = "red")    #  
X2 = np.array(x_2)                       # X  
Y2 = np.array(y_2)                       # Y  
plt.scatter(X2,Y2,s=25,color = "green")#  
  
clf = LinearDiscriminantAnalysis()        #      (  
clf.fit(X,Y)                              #  
print(clf.coef )                         #
```

```

print(clf.intercept_)          #
x = np.linspace(1.5,5,100)
#####
y = -1*(5.06950532*x-32.96135661)/6.28594264 #
plt.plot(x,y)
plt.scatter([1,4],[3,4],s=25,color = "blue")
plt.show
print(clf.predict([[1,4]]))
print(clf.predict([[3,4]]))

```

```

[[5.06950532 6.28594264]]
[-32.96135661]
[-1]
[1]

```

